

Readability vs. Faithfulness: Unveiling Correlations between Graph Aesthetics and DR Quality

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Abstract

Graph drawing aims to create a readable, faithful 2D graph layout, and dimensionality reduction aims to preserve the faithfulness of the layout. While an effective layout must simultaneously satisfy both readability and faithfulness, the lack of a statistical understanding of their relationship prevents the development of unified optimization strategies. Through a systematic study of 627 graphs, we evaluate the correlation between graph aesthetic and DR quality metrics and their robustness across graph size and density. Our results show that aesthetic metrics can be classified into four distinct groups based on their correlations with DR metrics, revealing highly complementary relationships. Moreover, specific metrics demonstrate strong, robust correlations that are robust to variations in graph scale. These findings suggest the necessity for a multi-faceted approach to layout evaluation and demonstrate that DR metrics can serve as computationally efficient proxies. By further validating their generalizability and scalability, future work can leverage this for multi-objective layout optimization, balancing readability and faithfulness.

CCS Concepts

• Human-centered computing → Graph drawings; • Mathematics of computing → Dimensionality reduction;

1. Introduction

Graph Drawing (GD) investigates the synthesis of readable 2D geometric representations that faithfully reflect the underlying topology of abstract data structures [PAV25]. While various criteria exist for effective graph drawing, aesthetic appeal, which focuses on readability, has traditionally been considered a priority in the graph drawing domain [PUR02]. A diverse range of aesthetic metrics serves to quantify how humans perceive the readability of a graph from various perspectives [PUR02, MHW*25], and algorithms are designed to enhance readability based on them [FR91, KK*89].

Another critical factor in graph drawing is faithfulness, how accurately a visual representation preserves the properties and relationships of the original abstract graph data [NEH13]. Fundamentally, 2D Euclidean space is not sufficient to fully embed graph theoretical distances of the original graph data [HK02]. By framing graph drawing as the projection of high-dimensional distances onto a 2D plane, dimensionality reduction (DR) techniques [EMK*19, JCJ*23] can be utilized to evaluate the faithfulness of a layout [PAV25].

Although established quantitative methodologies exist for measuring readability and faithfulness, the relationship between these

two aspects remains largely unexplored. While certain indicators, such as Stress, are employed in both domains [MHW*25, EMK*19], a single metric cannot fully capture the multifaceted nature of graph drawing and dimensionality reduction [vWMT25].

In this study, we investigate the relationship between graph aesthetic metrics [MHW*25] and DR quality metrics (or DR metrics) [JCJ*23] to identify layout strategies that are both readable and faithful. First, we define the ground truth as the layout generated by the MDS-initialized Kamada-Kawai algorithm [KK*89, GKN05] in a high-dimensional space that preserves 99% of the graph's information, and the projection as the 2D layout produced identically. Then we derive the DR quality metrics between the ground truth and the projection, and aesthetic metrics from the projection. Finally, we calculate the correlation between them.

Our analysis categorizes graph aesthetics into four functional groups based on their correlation with DR quality metrics. We identify universal metrics aligning across all DR dimensions, alongside complementary aesthetics linked to specific DR subsets. Furthermore, Aspect Ratio exhibits a robust inverse correlation with DR metrics, whereas Edge Orthogonality remains structurally independent. Consequently, our findings establish that specific DR metrics can serve as computationally efficient, differentiable surrogates for complex aesthetics. By addressing the need for edge-aware DR metrics and ensuring broad scalability and generalizability, future

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research can leverage DR-driven, multi-objective layout optimization to bridge the gap between readability and faithfulness.

2. Background and Related Work

This study bridges the aesthetic quality for readable graph drawing and the faithfulness measurements with DR techniques.

2.1. Graph Drawing for Aesthetics

The primary goal of traditional graph drawing is to produce readable and aesthetically pleasing layouts that minimize the viewer's cognitive load [GFV13]. To achieve this, force-directed algorithms are most prevalent. The Fruchterman-Reingold (FR) algorithm utilizes local attraction and repulsion to emphasize cluster structures and uniform node distribution [FR91] while the Kamada-Kawai (KK) algorithm optimizes the shortest-path distances between nodes to preserve global symmetry and the overall graph shape [KK*89]. Modern research continues to refine these forces to better capture complex structures of the original graph [FHM25].

To quantify visual quality, we utilize the comprehensive set of aesthetic metrics established by Mooney et al. [MHW*25]. Metrics such as Kruskal's stress, which are related to structural faithfulness rather than visual aesthetics, are excluded. Regarding node-related measures, **Node Uniformity** evaluates the evenness of node distribution across the drawing space, while **Node Resolution** assesses the distinguishability of nodes by calculating the ratio between the minimum and maximum distances between them.

Edge-related metrics focus on reducing visual confusion through geometric constraints. **Angular Resolution** and **Crossing Angle** measure the deviation from ideal angular distributions and 90° intersections, respectively, to ensure distinct edge perception. **Edge Orthogonality** and **Edge Length Deviation** assess the alignment with axes and the consistency of edge lengths, contributing to visual balance. Most critically, **Edge Crossings** quantifies the reduction of intersections, a key factor for readability.

Encompassing node- and edge-based variants, **Gabriel Ratio** evaluates proximity-based separation by measuring the extent to which the layout satisfies the Gabriel graph property. Finally, the **Aspect Ratio** evaluates the proportions of the layout's bounding box considering the entire geometry of nodes and edges.

2.2. Dimensionality Reduction for Faithfulness

From an information visualization perspective, a graph layout is considered informative if it accurately conveys the data's intrinsic topology, clusters, and neighborhood relationships [NEH13]. This perspective treats the graph drawing problem as the projection of high-dimensional features into a 2D space, where preserving relational distances and proximity is paramount [PAv25]. Consequently, various studies suggest algorithms that actively incorporate DR principles to ensure faithfulness [ZCH*21, FHM25].

To evaluate the faithfulness of projections, various metrics quantify the distortion or fidelity between the high-dimensional data X and its projection Y [EMK*19]. These metrics are broadly categorized into local, cluster-level, and global metrics based on their target structural granularity. Local metrics evaluate the preservation of

k -nearest neighbor (k NN) structures. **Trustworthiness** and **Continuity**, **Local Continuity Meta-Criteria (LCMC)**, and **Neighbor Dissimilarity (ND)** assess the overlap and consistency of neighborhood sets between X and Y . To further evaluate rank-based precision, the **Mean Relative Rank Error (MRRE)** and **Topographic Product** measure fluctuations in neighborhood distance rankings and topological consistency.

Cluster-level metrics [JKJ*22] evaluate the preservation of grouping structures and density distributions. **Steadiness** and **Co-hesiveness** assess whether clusters are unnaturally merged or split in the 2D plane, while **Distance-to-Measure** evaluates structural fidelity with robustness to local noise. Finally, global metrics quantify the distortion of the entire dataset's geometry. **Stress** and its variants, including *Non-metric* and *Scale-normalized* versions, measure the absolute or relative discrepancy in point-pairwise distances. Correlation-based **Pearson's r** and **Spearman's ρ** evaluate linear and rank-based distance alignment, while **Procrustes Measure** and **KL Divergence** assess global shape similarity and probability distribution consistency, respectively.

3. Methodology

In this study, we adopt a quantitative approach to analyze the relationship between readability and faithfulness in graph drawing. To ensure generalizability across diverse visualization scenarios, we utilize the dataset from Seokweon et al. [JRD*26], which comprises 627 synthetic and real-world graphs with varying scales ($|V| \in [10, 400]$) and densities ($|E|/|V| \in [1, 10]$).

We employ the Kamada-Kawai (KK) algorithm [KK*89] initialized with Multidimensional Scaling (MDS) [GKN05] in generating both a 2-D layout and its ground truth in high-dimensional space, to ensure a balance of **Readability**, **Faithfulness**, and **Reproducibility**. Force-directed algorithms like KK inherently adhere to aesthetic principles, generally producing more readable layouts than DR-based techniques [GFV13, KRM*17]. Furthermore, KK directly promotes structural faithfulness through its objective function, which minimizes stress by reducing the discrepancy between graph-theoretic shortest path distance and low-dimensional Euclidean distances [KK*89]. Finally, by utilizing MDS as the initial seed, we mitigate the risk of falling into local minima while guaranteeing deterministic and reproducible results [GKN05].

As high-dimensional spaces offer greater capacity to preserve the information contained in a graph's distance matrix without distortion [HK02], we regard the layout derived from the MDS-initialized KK algorithm in a sufficiently high-dimensional space as the ground truth (GT). To determine the appropriate dimensionality for each graph, we adopt the Isomap algorithm to calculate the intrinsic dimension, N , which is the minimum number of dimensions required to preserve at least 99% of the variance inherent in the graph's original distance matrix [TSL00]. The standard 2D layout with the same algorithm is regarded as a projection of GT.

We leverage existing implementations for metrics detailed in section 2: the ZADU library for DR quality [JCJ*23] and the implementation by Mooney et al. for graph aesthetics [MHW*25]. To ensure directional consistency, we transform all metrics so that higher values represent improved faithfulness and readability. We

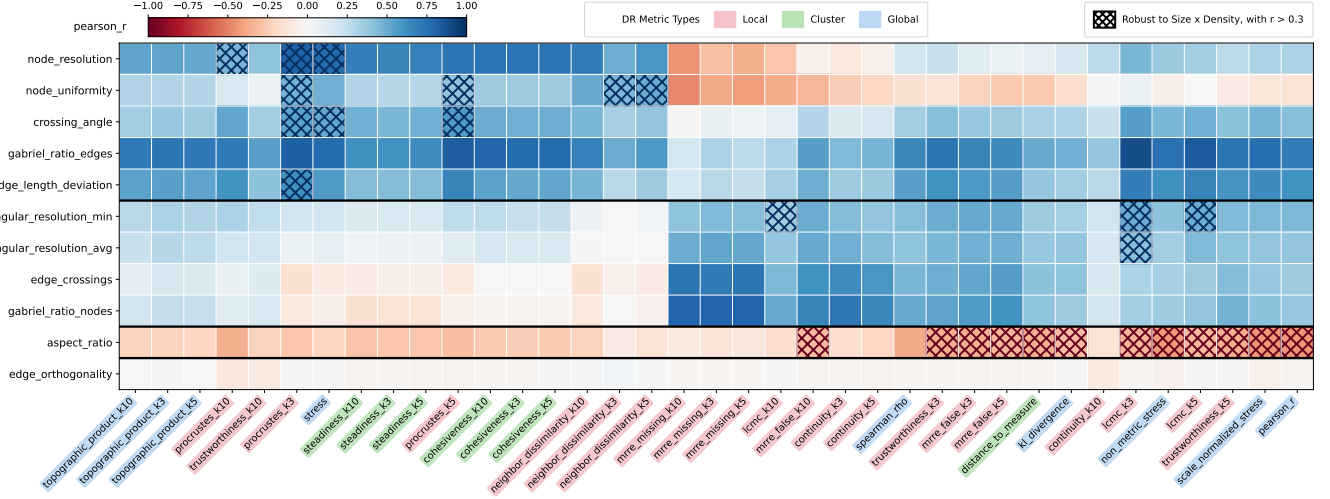


Figure 1: Heatmap of Pearson’s r correlations between aesthetic (Rows) and DR metrics (Columns). All metrics are transformed so that higher values represent better faithfulness and readability. Bold horizontal lines group aesthetic metrics by similar patterns, while DR metrics are grouped by granularity level. Cells with $(|r| > 0.3)$, where the relationship remains robust against variations in graph size and density as verified by moderation analysis, are highlighted with a hatched pattern.

then compute Pearson’s r for each pair of metrics to investigate their statistical relationship. Additionally, we employ a moderation analysis to verify that the influence of DR metrics on aesthetic outcomes remains robust against variations in graph size and density.

4. Results

We systematically categorize the aesthetic metric into four distinct groups based on their correlation with DR metrics (see Figure 1).

Group 1: Topology Preserving Aesthetics (Rows 1-5): Gabriel Ratio-Edges, Crossing Angle, Edge Length Deviation, Node Resolution, and Node Uniformity exhibit strong, robust positive correlations with DR metrics assessing topological preservation across all levels of granularity. Notably, Procrustes ($k = 3$) maintains robust relationships with four out of five aesthetic metrics, demonstrating a particularly high correlation with Node Resolution ($r \approx 0.8064$). While Gabriel Ratio-Edges yields the highest average correlation overall ($r \approx 0.6198$) and an even stronger peak correlation, it is highly sensitive to variations in graph size and density.

Conversely, their correlations with local neighborhood-centric metrics such as MRRE and Continuity are relatively weak or even negative and fail to demonstrate robustness. Because these aesthetic metrics primarily evaluate how evenly and distinctly nodes are distributed in 2D space without overlapping, the original high-dimensional density becomes distorted, inevitably leading to the loss of true nearest neighbors.

Group 2: Local Neighborhood Preserving Aesthetics (Rows 6-9): On the contrary, this group, which consists of aesthetics related to edge separation, aligns strongly with local DR quality metrics such as Continuity and MRRE. This demonstrates that the degree

of clear edge separation in a layout is closely tied to the successful preservation of local neighborhood relationships. Specifically, while **Edge Crossing**, **Gabriel Ratio-Nodes** yield high baseline correlations (up to $r \approx 0.7953$) but lack robustness, **Angular Resolution (Avg, Min)** exhibits a distinctly robust correlation with LCMC, despite their relatively lower correlation ($r \approx 0.4944$).

Interestingly, the Gabriel Ratios present a naming paradox: **Gabriel Ratio-Edges** aligns with Node Distribution (Group 1), while **Gabriel Ratio-Nodes** aligns with Edge Clarity (Group 2). This indicates that the aesthetic properties these metrics actually measure are inextricably tied to their opposite structural features.

Group 3: Negative Aesthetics (Row 10): Surprisingly, **Aspect Ratio** exhibits a negative correlation to every graph DR metric, to be robust in size and density with DR metrics for local neighborhood preservation. This indicates that forcing a graph to fit within a square bounding box can actually be disadvantageous for preserving high-dimensional structural information.

Group 4: Independent Aesthetics (Row 11): Edge Orthogonality shows weak correlations ($|r| < 0.1$) across all DR metrics, functioning merely as stylistic rather than structural indicators.

5. Discussion

Our findings reveal that graph readability is deeply intertwined with structural faithfulness in a complementary manner. Therefore, evaluating layouts for both readability and faithfulness requires a multifaceted approach utilizing diverse metrics.

Furthermore, our analysis demonstrates that specific DR metrics can serve as efficient, differentiable surrogates that are robust against varying graph scales. Deriving traditional aesthetics from

graph data requires a two-stage computational bottleneck: generating the layout, then running complex geometric calculations. By substituting these steps with optimized DR metrics, the computational overhead can be drastically reduced.

Synthesizing these insights leads to DR-driven, multi-objective layout optimization. If we can identify DR metrics that strongly and robustly represent multi-faceted graph aesthetics, layout algorithms could seamlessly optimize for these metrics. This would facilitate the highly efficient generation of optimal graph layouts, leveraging a comprehensive multi-objective approach to explicitly balance the varied dimensions of both readability and structural faithfulness.

6. Limitations

To realize this vision, verifying the generalizability and scalability of our findings across diverse graph characteristics requires further investigation. Because our experiments center on the KK algorithm, different correlation patterns may emerge under alternative layouts. Additionally, our selected aesthetic and DR metrics should be further contextualized within the broader GD and DR literature, expanding upon current prior research [JCJ*23, MPWK24]. Moreover, there is a critical need to discover the relationships between DR and graph aesthetics across larger, more diverse graph scales than those currently tested. Addressing these limitations in future work is essential to rigorously validate the quantitative interplay between readability and faithfulness.

7. Conclusion

Through a systematic quantitative analysis, this study quantifies the alignment between graph aesthetics and DR quality metrics. Our results demonstrate that, although certain aesthetic metrics align well with most DR quality metrics, inherent trade-offs exist; therefore, these metrics must be used in a complementary manner to provide a holistic evaluation. These insights highlight that DR quality metrics can provide a computationally efficient framework for multi-objective optimization of graph layout, effectively bridging the gap between readability and faithfulness.

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